

Arya Verma

aryaverma0717@gmail.com | +1 (908) 230-4825 | New Providence, NJ 07974 | [LinkedIn](#) | [GitHub](#)

Education

University of California San Diego

B.S. Data Science, Economics

Coursework: Data Structures & Algorithms, Statistics, Linear Algebra, Calculus, AI/ML, Neural Nets, Economic Principles and History.

Expected Graduation March 2027

GPA 3.76/4.0 (Provost Honors)

Experience

UCSD Economics Department

Research Assistant

Sep – Dec 2025

San Diego, CA

- Built a GIS dataset of 600,000 Indian settlements to model urban sprawl and expansion as affected by land ownership restrictions.
- Clustered 375 metropolitan areas with GeoPandas, NumPy, and scikit-learn to track polycentric growth and regional spillover effects.
- Analyzed 20 years of emissions data from Google Earth Engine, georeferenced official maps to uncover policy-driven disparities.

QuaEra Insights

Data Science Intern

Jun – Aug 2025

New Providence, NJ

- Rewrote legacy code and documentation to create a scalable data preprocessing pipeline, improving reliability and consistency.
- Engineered 85+ predictive features (biographical, behavioral, semantic, temporal) from 2.8M+ healthcare marketing records using PySpark SQL, Word2Vec embeddings, and LLM-based extraction, boosting accuracy in engagement-model predictions.
- Presented schema and preliminary engagement trends, allowing for reevaluation of previous projects and client strategy optimization.

RCG Global Services

Data Science Intern

Aug – Sep 2024

Iselin, NJ

- Developed an ETL pipeline to parse and structure 100+ project tasks from Trello boards into a consolidated, queryable dataset.
- Utilized Python, Pandas, and SQL to transform data, integrate AI models, construct a relational database, and automate Excel exports.
- Streamlined project management workflows, allowed for instantaneous consolidation, and improved data storage and accessibility.

Projects

Neural U.S. President Speech Classification

- Implemented a GPT-style Transformer in PyTorch with multi-head self-attention, incorporating causal masking, layer normalization.
- Evaluated performance through speech classification (80%+ accuracy) and autoregressive language modeling with perplexity analysis.

KoreAI [\[GitHub\]](#) | *Best Presentation at UCSD DataHacks 2025, 40+ teams*

- Developed an ML tool to predict optimal clothing colors for users from facial RGB data extracted with OpenCV and MediaPipe.
- Trained a KNN classifier on a custom dataset of 100+ celebrities, achieving 85% test accuracy and real-time functionality.
- Demonstrated product features and commercial applications to an audience of 150+ students, faculty, and industry professionals.

SMASH & NSF HDR Sea Anomaly Detection [\[GitHub\]](#)

- Extracted and modeled 20 years of sea level data from 7,300+ tide gauge files to predict flood risk across 12 East Coast stations.
- Trained a Random Forest Classifier with 97.7% test accuracy to improve proactive flood planning through 10-year future predictions.
- Detected anomalies through time series modeling of engineered features and identified irregular coastal patterns to preempt flooding.

Leadership

UCSD Computer Science and Engineering Society

Product Manager

Jul 2025 – Present

San Diego, CA

- Determine user base and needs, design and plan features for Opportune, a career growth and networking platform for UCSD students.
- Set timetables, organize workflows and cross-functional collaboration, create marketing opportunities, and assess product feedback.

Skills and Interests

Programming languages: Python, SQL, MATLAB, R, Java, JavaScript, HTML, CSS, Stata

Libraries & Tools: Pandas, NumPy, Scikit-learn, Apache (Spark, Hadoop), PyTorch, Matplotlib, GeoPandas, TensorFlow, D3.js

Software and platforms: DataBricks, Microsoft Excel, PowerPoint, QGIS, Power BI, AWS, Azure, Git, Jupyter Notebook

Machine learning: Viterbi, EM, Gradient Descent Algorithms; Linear, Logistic, Decision Tree, SVM Models, Fourier Transforms

Interests: Linguistics, NLP, world languages, history, applied economics, guitar, athletics, traveling

Awards

Second Place at the 2024 SDSU Big Data Hackathon: Competed with 60+ teams at the Sharp HealthCare-sponsored event.

Gold Award in Level 2 AATG National German Exam: Awarded for scores in the 90th percentile or above.